

Chapter 5.A

Stoichiometry

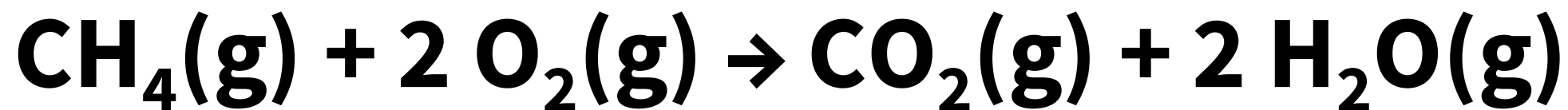
Chemical Principles I

CHEM 1211K

1. Calculate the number of moles of any substance involved in a chemical reaction from the number of moles of any other substance in the reaction.
2. Use the mass of one substance to determine the masses of other substances involved in a chemical reaction.
3. Calculate the quantities of substances produced in a reaction when quantities of more than one reactant are specified.
4. Express the quantity of product obtained from a reaction as a percentage of what the reaction is theoretically capable of producing.
5. Solve problems using molarity.
6. Extend the concept of molarity to the individual ions in solutions of ionic substances.

- **Stoichiometry** is the study of quantitative relationships involving mass and number of moles in chemistry.
- Any chemical question that asks “how much?” of a particular substance is consumed or formed in a chemical reaction is a stoichiometry problem.

- **Molar ratios (mole-mole ratios)** derived from stoichiometric coefficients in a balanced chemical equation are used in stoichiometry problems to “convert” from amount of one substance to amount of another.



How many moles of O₂ are required to produce 12.5 mol of NO₂ according to the balanced chemical equation below?



5.2 Mass Calculations for Chemical Reactions | 1 of 2

- Moles cannot be measured directly—in the laboratory, mass and volume measurements are typically used to quantify substances.
- **Molar mass (g/mol)** enables us to determine moles from mass or *vice versa*.

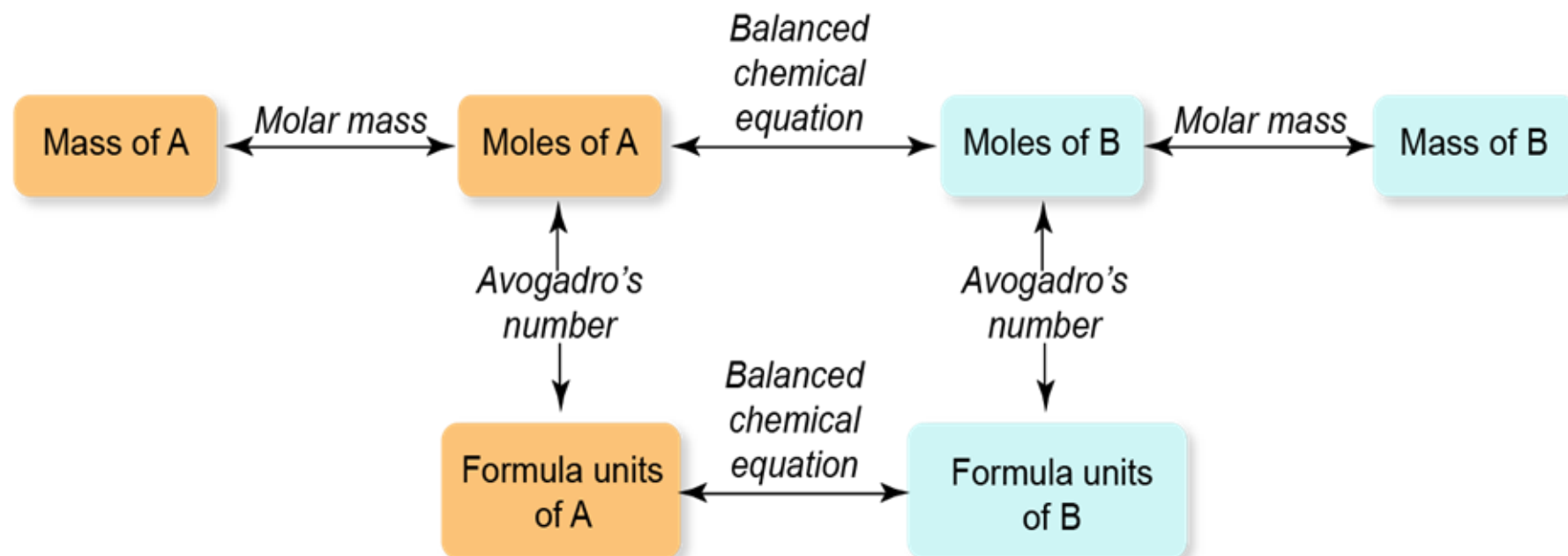


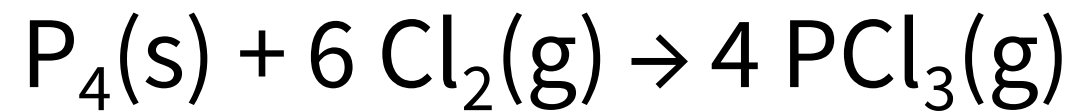
Figure 5.2. Flow chart for stoichiometry problems involving mass.

- When working a stoichiometry problem, creating a “solution flowchart” *before* plugging in numbers is highly recommended.
- **Example.** What mass of oxygen gas (O_2) from the air is consumed in the combustion of 702 g of octane (C_8H_{18})?

How many grams O_2 are required to produce 27.5 g of NO_2 according to the balanced chemical equation below?



What mass of Cl_2 is required to produce 332 g PCl_3 in the presence of **excess** P_4 ?



- The exact amount of a substance needed to consume another substance is called the **stoichiometric amount**.
- Reactants are frequently not combined in stoichiometric amounts; in this case, one reactant will run out before the others.
 - The reactant that is fully consumed is called the **limiting reactant**.
 - Reactant(s) that remain at the end of the reaction are called **excess**.

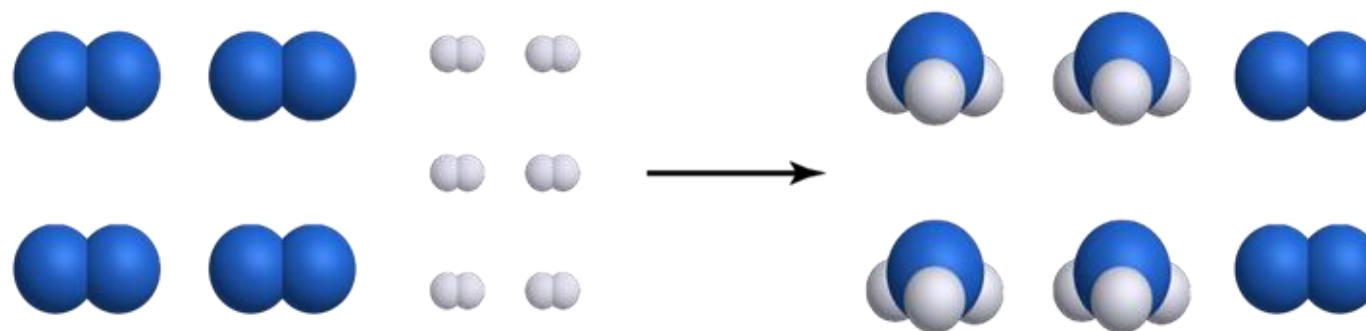
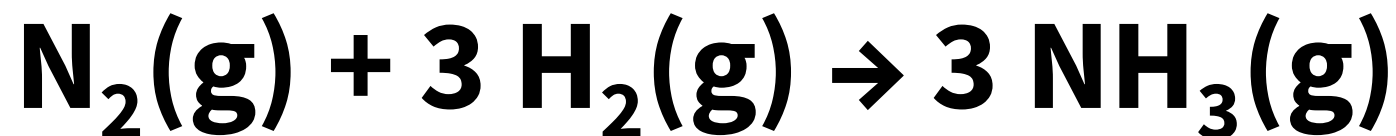


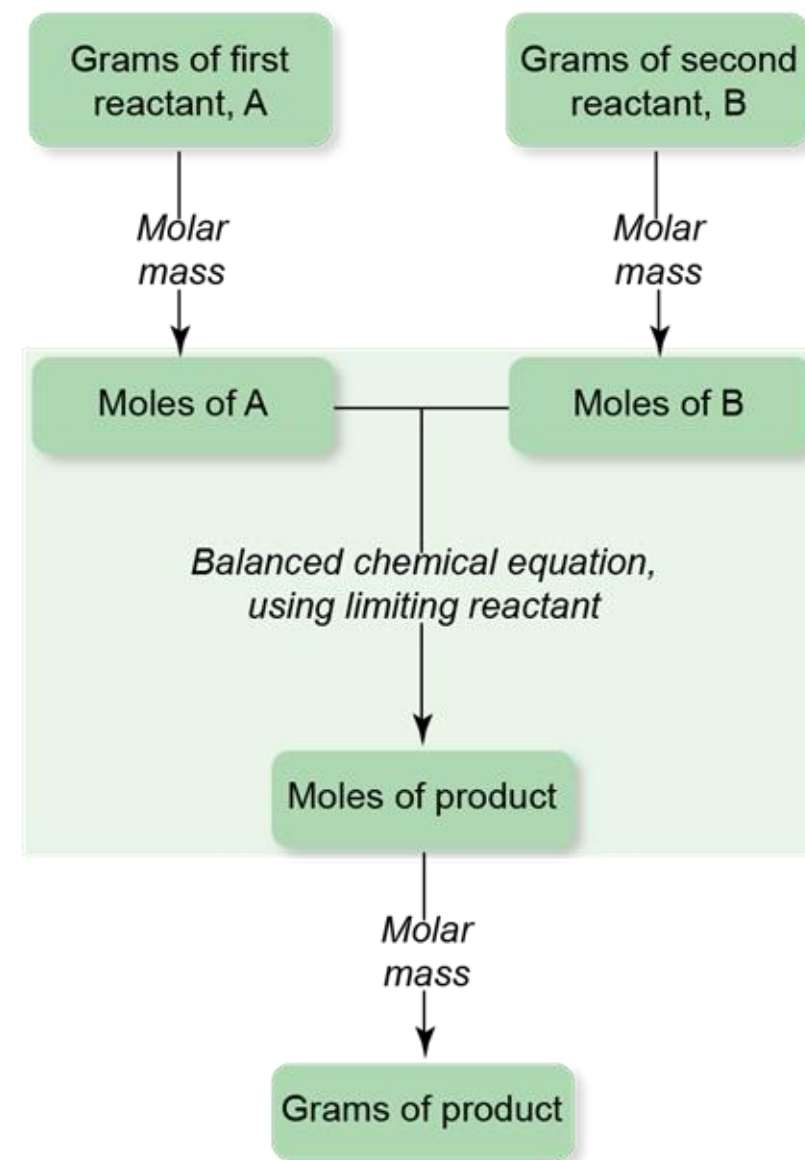
Figure 5.4. Reaction between nitrogen (blue) and hydrogen (white) to produce ammonia.

- To better understand the concept of a limiting reactant...let's make some sandwiches!
- Given the recipe below, how many sandwiches can we make with 26 bread, 20 ham, and 10 cheese?

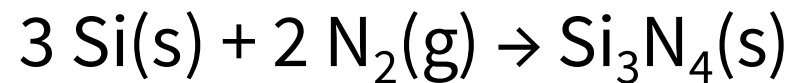
2 bread + 4 ham + 2 cheese → 1 sandwich

5.3 Problems Involving Limiting Quantities | 3 of 4

- The limiting reactant is the substance capable of producing the *smallest* amount of product (or reaction events).
- The theoretical mass of product is determined *only* from the moles of the limiting reactant.
- Amounts of excess reactant(s) remaining are determined by subtracting the amount of reactant consumed from the initial amount.
- **Think about sandwiches!**



Example. Silicon nitride is prepared according to the following equation:



Which reactant is limiting when 2.00 g of Si and 1.50 g of N₂ react?

Chapter 5.B

Stoichiometry

Chemical Principles I

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5.3 Formal Determination of Theoretical Yield

- Formally, consider a reaction with M reactants R_i (with coefficients r_i) and N products P_j (with coefficients p_j), with chemical equation



- Given an initial amount in moles $n(R_i)$ for each reactant, the limiting reactant is the substance for which $n(R_i)/r_i$ is smallest. This ratio is the **number of reaction events** x_i that each reactant can carry out; the smallest x_i is the limiting number of reaction events x_{lim} .

$$x_{lim} = \min [n(R_1)/r_1, n(R_2)/r_2, \dots, n(R_N)/r_N]$$

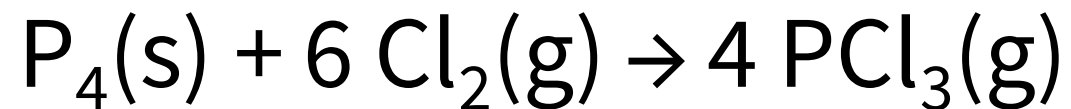
- For any product P_j , we can calculate the **theoretical yield** in moles using

$$n(P_j) = p_j \cdot x_{lim}$$

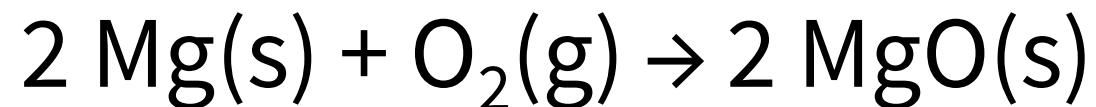
- For each reactant that is not limiting, the excess remaining is the following, where *fin* and *init* refer to “final” and “initial” numbers of moles respectively.

$$n(R_i)_{fin} = n(R_i)_{init} - r_i \cdot x_{lim}$$

What is the maximum mass (in g) of PCl_3 that can be produced from the reaction of 125 g P_4 with 323 g Cl_2 ? Enter your answer without units.

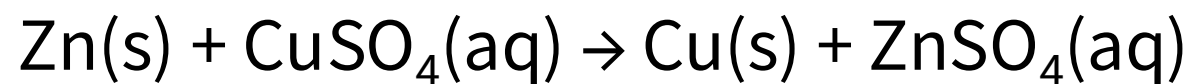


If 42.5 g of O₂ reacts with 75.1 g of Mg to produce MgO, then what mass of excess reagent remains?
Assume 100% yield for the reaction.

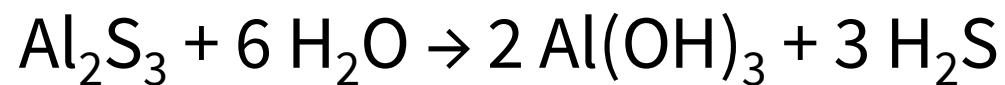


- **Theoretical yield** is the hypothetical maximum mass or moles of product given conservation of mass and a balanced chemical equation.
- **Actual yield** is the measured mass or moles of product obtained.
- **Percent yield**

Example. Upon reaction of 1.274 g of copper(II) sulfate with excess zinc metal, 0.392 g copper metal was obtained according to the equation below. What is the percent yield of copper metal?



A 316.0 g sample of aluminum sulfide reacts with 493.0 g of water to produce aluminum hydroxide and dihydrogen sulfide. Determine mass of aluminum hydroxide that forms if the percentage yield for the reaction is 85.0%.



- Recall that in an **aqueous solution**, a **solute** is dissolved in water (**solvent**).
- Concentration** is a measure of the amount of solute in a fixed amount of solution.

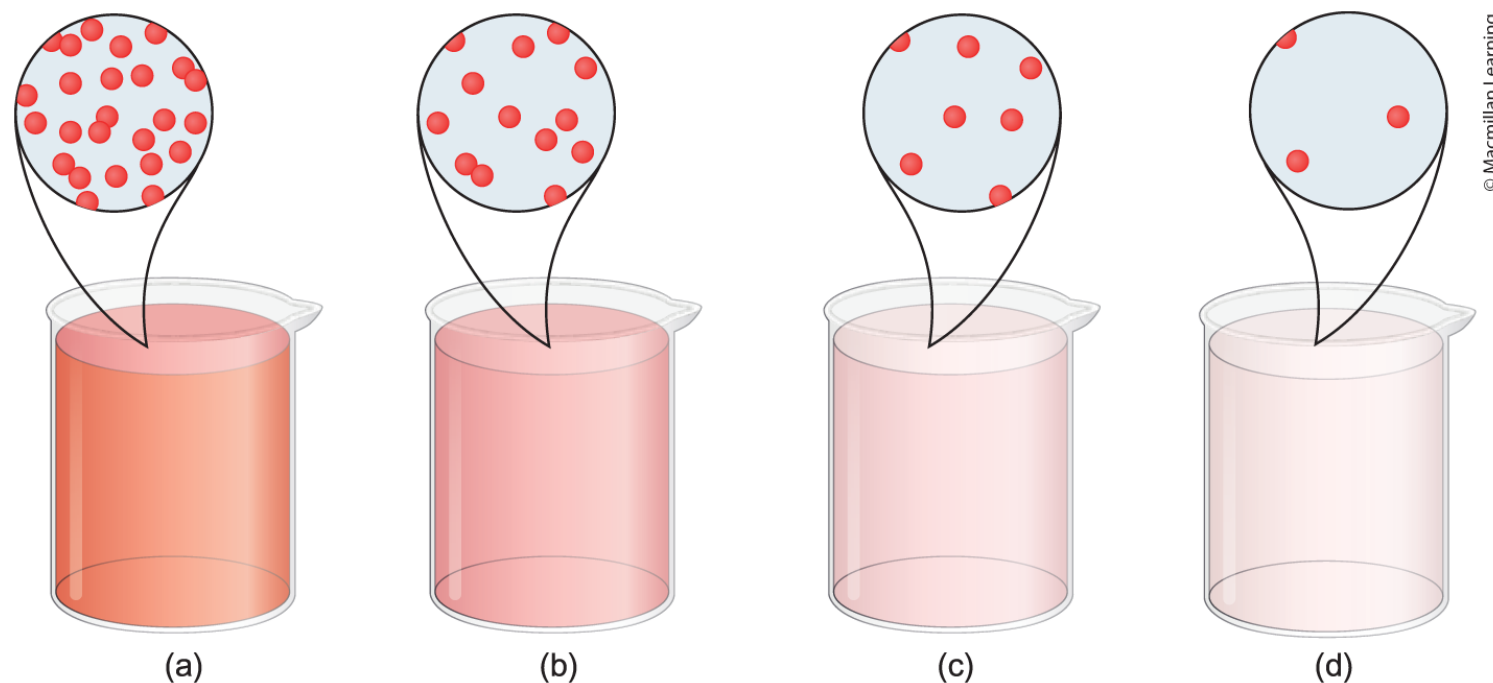


Figure 5.8. A series of solutions from concentrated (a) to dilute (d).

- The most important expression of concentration in chemistry is **molarity (mol/L or M)**.

Chapter 5.C

Stoichiometry

Chemical Principles I

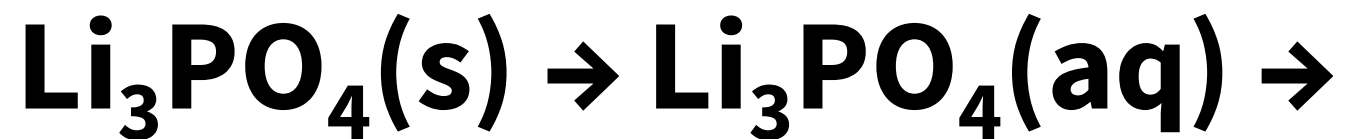
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5.6 Molarities of Ions



- Soluble ionic compounds **dissociate completely** into ions when dissolved in water. Formula units of ionic compounds may contain more than one ion!
- The “molarity of an ion” is the number of moles of that ion per liter of solution.
- Subscripts on monatomic or polyatomic ions indicate the molar ratio of that ion to the formula unit as a whole.



What is the molarity of calcium ions in a solution made by dissolving 23.26 g $\text{Ca}_3(\text{PO}_4)_2$ (0.07500 mol) in enough water to produce 200.0 mL of solution? Enter your response to the nearest 0.001 M.

What mass of Na_3PO_4 is required to produce 500.0 mL of solution that is 0.750 M in *sodium ions*?

The molar mass of Na_3PO_4 is 163.94 g/mol and that of Na^+ is 22.99 g/mol. Enter your response to the nearest 0.1 g.

- **Dilution** is the process of adding pure solvent to a solution, lowering the concentrations of all solutes.

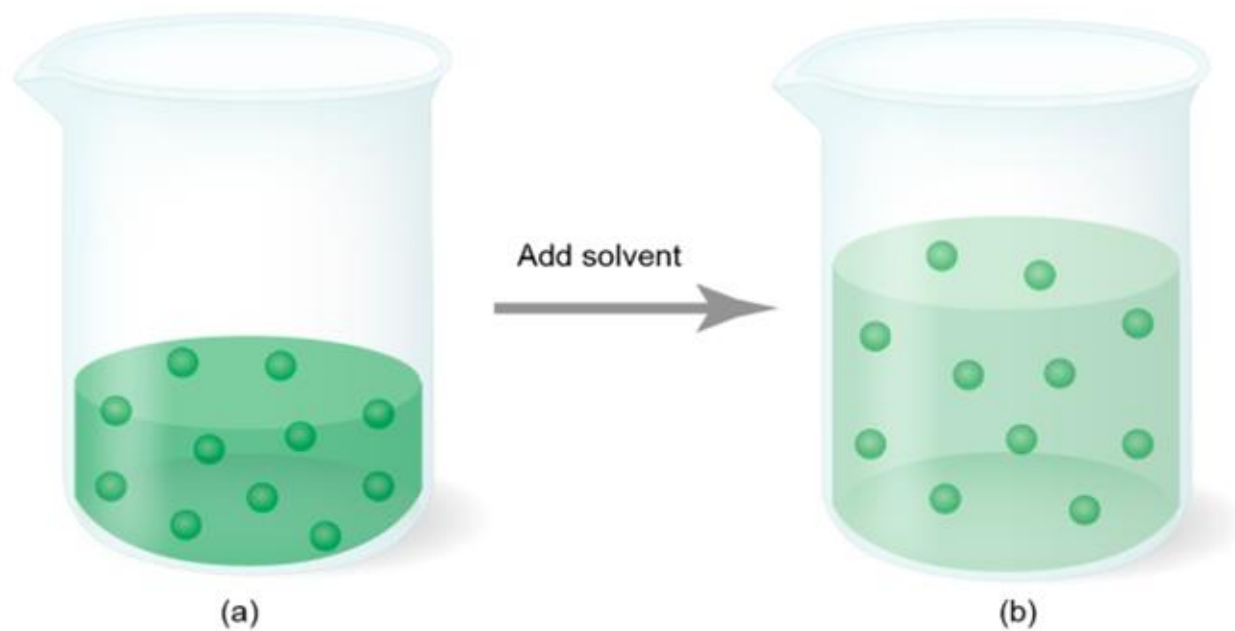


Figure 5.11. Adding solvent dilutes solution (a) to solution (b).

Example. When a 1.25 M solution of NaCl in water is diluted from 25.0 mL to 100. mL, what is the concentration in mol/L of the resulting solution?

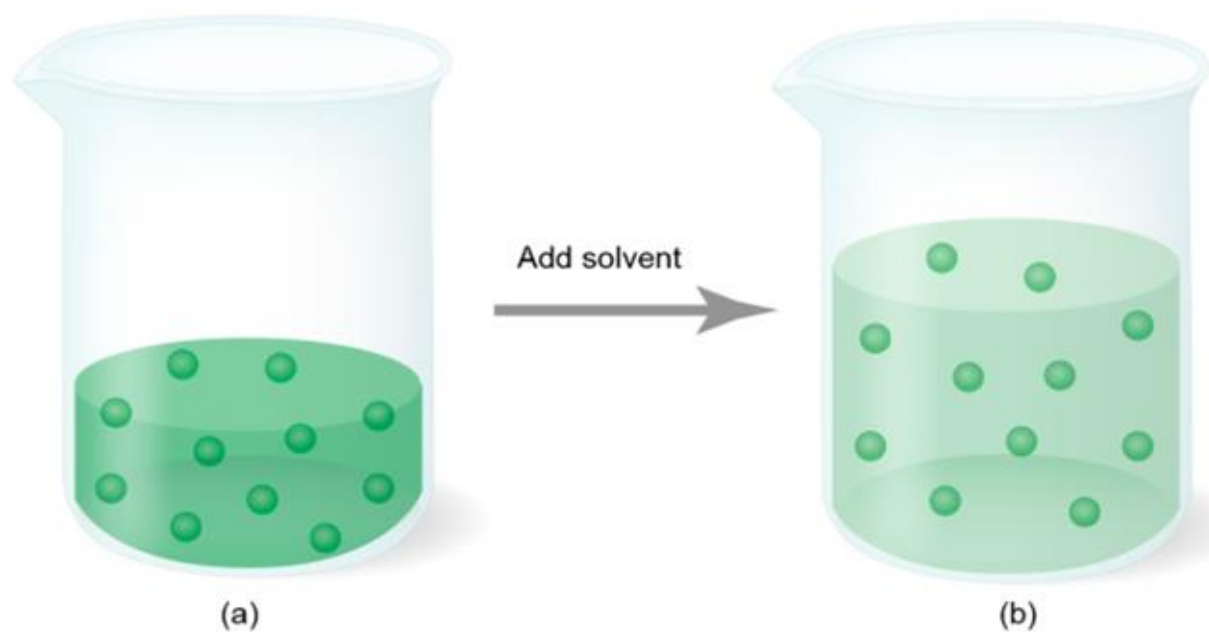


Figure 5.11. Adding solvent dilutes solution (a) to solution (b).

- **Solution stoichiometry** is the application of stoichiometry to reactants and/or products in solution.
- Molarity is another tool in the “stoichiometry toolbox” enabling us to determine moles of solute from volume of solution or *vice versa*.

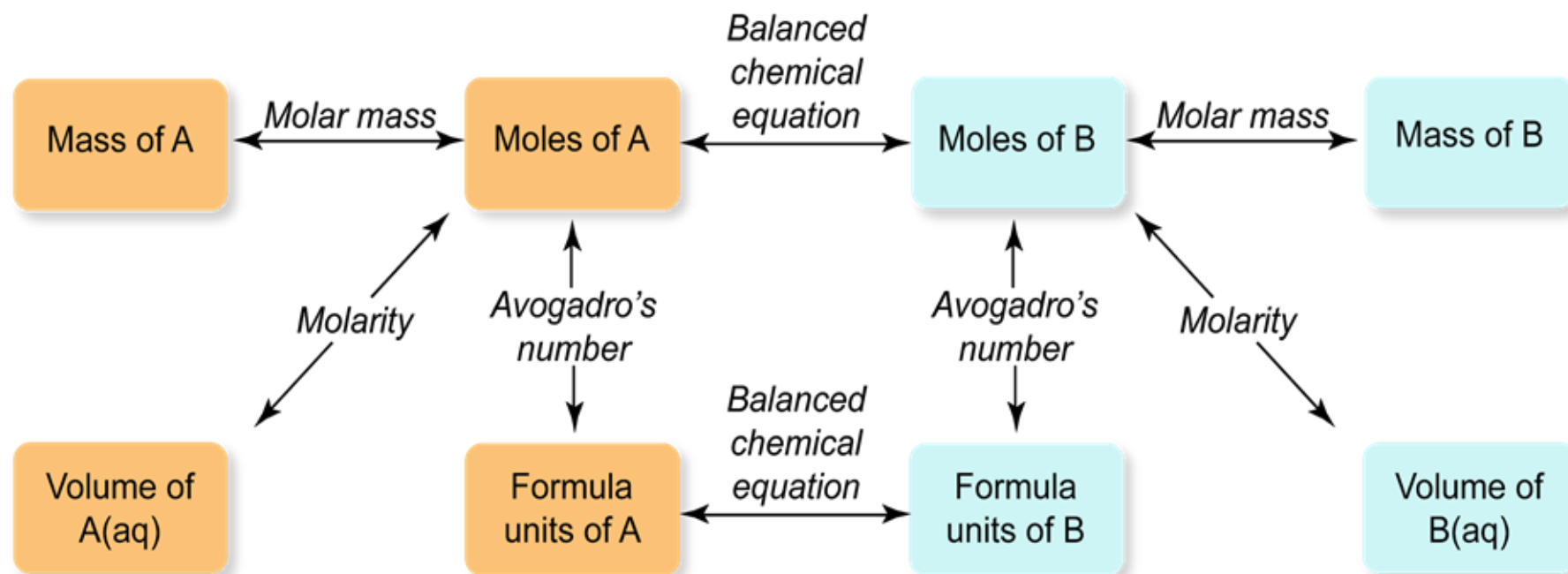
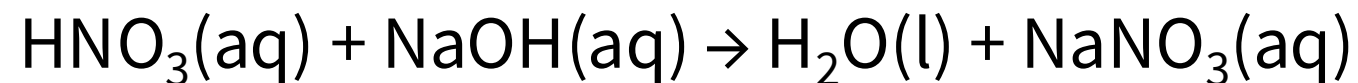


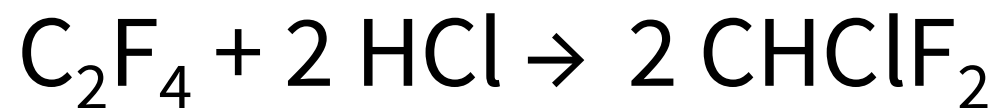
Figure 5.14. Flow chart for stoichiometry problems incorporating aqueous solutions.

A 20.00 mL sample of aqueous HNO_3 reacts with 0.110 M NaOH. Determine the concentration of HNO_3 if 49.22 mL of NaOH is required for complete reaction (with 100% yield). Enter your response to the nearest 0.001 M.



- Our “stoichiometry toolbox” now contains several ways to determine moles from measured quantities...
 - **Molar mass:** mass → moles
 - **Molarity:** solution volume → moles of solute
 - **Density and molar mass:** substance volume → moles
 - **Avogadro's number:** number of particles → moles
- Note that we will add the ideal gas law ($n = PV/RT$) to this list later in the course.

Example. Calculate the moles of C_2F_4 needed to produce 185 mL of CHClF_2 .
The density of CHClF_2 is 1.49 g/mL.



- Some of the amazing things we can now do with stoichiometry...
 1. Extract molar ratios from a balanced chemical equation and use them to relate amounts of reactants and/or product to one another.
 2. Relate masses of reactants and/or products to one another stoichiometrically using molar ratios and molar masses.
 3. Determine limiting reactant given a balanced chemical equation and amounts or masses of all reactants.
 4. Calculate theoretical yield of a given product using a balanced chemical equation and amounts or masses of all reactants.
 5. Calculate or apply percent yield given an actual yield and sufficient information to determine theoretical yield.
 6. Determine or apply the molarity of a given solute in an aqueous solution; apply molarity in stoichiometric calculations for reactions in solution.
 7. Apply the dilution equation to plan or assess a dilution operation.